

Math 340 Final Exam Information and Review

- Final Exam date, time, and room information are posted on Canvas. It will be cumulative, covering everything on Midterms 1 and 2 AND Chapters 42-46, 36-37 in your textbook.
- Note: You will not be expected to do the Gram-Schmidt orthogonalization process on the Final. However, you are expected to know how to take a given orthogonal basis and create an orthonormal basis.
- The Final Exam is a 120-minute exam consisting of True/False questions, multiple choice problems, and free response questions.
- The only permitted materials are: pencil/pen. Any other materials/resources are prohibited.
- This review is intended to give you extra practice with connecting the concepts covered in the new sections. The problems on the exam are not guaranteed to be similar to these problems. Combine Midterms 1 and 2 Reviews with this review to review for the final.
- For additional practice, I recommend going through your homework problems, discussion worksheets, and Lecture examples.

New terms to know: matrix transformation, linear transformation, domain space, codomain space, linearity properties, image vector, linear operator, standard matrix of a linear transformation, zero transformation, zero operator, identity operator, scalar operator, coordinate transformation, kernel of a linear transformation, nullity, range/image of a linear transformation, rank of a linear transformation, composite function, injective/one-to-one function, surjective/onto function, isomorphism, isomorphic vector spaces, inverse of a linear transformation, matrix of a linear transformation corresponding to ordered bases, eigenvalue/eigenvector/eigenspace of a linear operator, diagonalizable linear operator, inner product, inner product space, norm, unit vector, distance, angle, orthogonal vectors, orthogonal set, orthonormal set

New Learning Goals for Final (In addition to Midterm 1 and 2 Learning Goals)

Clicking on each section will refer you to corresponding page in the review packet.

- Be able to show a function between vector spaces is or is NOT a linear transformation
- Find the standard matrix representation of a linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$
- Know how to find the kernel and/or range/image of a linear transformation (as well as a basis for the kernel/image)
- Know how to determine if a linear transformation is one-to-one and/or onto
- Know the dimension theorem and how to use it to calculate possible dimensions of kernel and image.
- Know how the dimension of vector spaces for a linear transformation interacts with the problem of being one-to-one/onto
- Be able to determine if a linear transformation is an isomorphism (both using the definition and using a matrix representation)

- Know how to determine if two finite-dimensional vector spaces are isomorphic
- Know how to find the matrix representing a linear transformation $T : V \rightarrow W$ using ordered bases for vector spaces V and W
- Be able to find kernel/image of a linear transformation using a matrix representation
- Know how to change the representation of a linear operator $T : V \rightarrow V$ from one basis to another, both directly and using the Similarity Theorem
- Be able to find the eigenvalues/eigenvectors of a linear operator
- Be able to find a diagonal matrix representation of a linear operator (if possible)
- Know how to verify if a function is an inner product
- Be able to compute the length/norm of a vector and find a unit vector given an inner product
- Be able to find the distance between two vectors using an inner product
- Know how to check if two vectors are orthogonal given an inner product
- Be familiar with the Cauchy-Bunyakovsky-Schwarz inequality and its implications
- Be able to build an orthonormal basis given an orthogonal basis
- Know how to find the coefficients of a linear combination using an orthogonal/orthonormal basis
- Be familiar with the list of invertibility criteria (Box of Facts)

Explanations and Example Problems

- Be able to show a function between vector spaces is or is NOT a linear transformation

Definition: Let V and W be vector spaces. A function $T: V \rightarrow W$ is called a linear transformation of V into W if

- (a) $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for every $\mathbf{u}, \mathbf{v}$ in V .
- (b) $T(c\mathbf{u}) = cT(\mathbf{u})$ for every $\mathbf{u}$ in V and c in $\mathbb{R}$.

Warning: In the above definition, $\mathbf{u} + \mathbf{v}$ is addition happening in V , while $T(\mathbf{u}) + T(\mathbf{v})$ is addition happening in W .

Example: The function $L: M_{22} \rightarrow \mathbb{R}$ by $L(A) = \text{tr}(A)$ is a linear transformation, while $D: M_{22} \rightarrow \mathbb{R}$ by $D(A) = \det(A)$ is NOT a linear transformation.

Problem: Show that the function $T: P_1 \rightarrow P_2$ by $T(p(x)) = xp(x)$ is a linear transformation, while $R: P_1 \rightarrow P_2$ by $R(p(x)) = p(x) + x^2$ is NOT a linear transformation.

Solution: To show T is a linear transformation, we need to show (1) $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ and $T(c\mathbf{u}) = cT(\mathbf{u})$ for every $\mathbf{u}, \mathbf{v}$ in V and c in $\mathbb{R}$.

Let $p(x), q(x)$ be in P_1 , and c in $\mathbb{R}$. Then

$$(1) T(p(x) + q(x)) = x(p(x) + q(x)) = xp(x) + xq(x) = T(p(x)) + T(q(x)).$$

Also,

$$(2) T(cp(x)) = x(cp(x)) = c(xp(x)) = cT(p(x)).$$

To show $R(p(x)) = p(x) + x^2$ is not a linear transformation, we need to show either that (1) fails (for an explicit pair of polynomials) or (2) fails (for explicit polynomial and scalar).

Showing (1) fails: Consider $p(x) = x + 2, q(x) = 2x + 3$.

$R(p(x) + q(x)) = R(3x + 5) = 3x + 5 + x^2$. However, $R(p(x)) + R(q(x)) = (x + 2 + x^2) + (2x + 3 + x^2) = 3x + 5 + 2x^2$. These are not the same polynomials, so $R(p(x) + q(x)) \neq R(p(x)) + R(q(x))$.

You could also show (2) fails: Consider $p(x) = x + 2, c = 2$.

Then $R(cp(x)) = R(2x + 4) = 2x + 4 + x^2$. However, $cR(p(x)) = 2(x + 2 + x^2) = 2x + 4 + 2x^2$. These are not the same polynomials, so $R(cp(x)) \neq cR(p(x))$.

- Find the standard matrix representation of a linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$

Tip: A function $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a matrix transformation exactly when it is a linear transformation. In this case, $T = T_A$ is the matrix transformation induced by the unique matrix A , given in terms of its columns by

$$A = [T(\mathbf{e}_1) \quad T(\mathbf{e}_2) \quad \cdots \quad T(\mathbf{e}_n)],$$

where $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ are the standard basis vectors of $\mathbb{R}^n$.

Problem: Let $L: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be a linear transformation defined by

$$L\left(\begin{bmatrix} 1 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} 3 \\ 5 \\ 7 \end{bmatrix}, \quad L\left(\begin{bmatrix} 2 \\ 4 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} \quad (1)$$

Find the standard matrix representing L .

Solution: Our goal is to find $L(\mathbf{e}_1)$ and $L(\mathbf{e}_2)$, both of which are vectors in $\mathbb{R}^3$. Note that we have

$$L\left(\begin{bmatrix} 1 \\ 3 \end{bmatrix}\right) = L(\mathbf{e}_1 + 3\mathbf{e}_2). \quad (2)$$

Because L is a linear transformation, we have

$$L(\mathbf{e}_1 + 3\mathbf{e}_2) = L(\mathbf{e}_1) + 3L(\mathbf{e}_2) = \begin{bmatrix} 3 \\ 5 \\ 7 \end{bmatrix} \quad (3)$$

Likewise, we have

$$L\left(\begin{bmatrix} 2 \\ 4 \end{bmatrix}\right) = L(2\mathbf{e}_1 + 4\mathbf{e}_2) = 2L(\mathbf{e}_1) + 4L(\mathbf{e}_2) = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}. \quad (4)$$

Subtracting 2 times the first equation by the second equation gives

$$2L(\mathbf{e}_2) = \begin{bmatrix} 4 \\ 6 \\ 8 \end{bmatrix} \quad (5)$$

which implies

$$L(\mathbf{e}_2) = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \text{ and } L(\mathbf{e}_1) = \begin{bmatrix} -3 \\ -4 \\ -5 \end{bmatrix}. \quad (6)$$

Hence, the standard matrix representing L , which is concatenating the vectors $L(\mathbf{e}_1)$ and $L(\mathbf{e}_2)$, is

$$\begin{bmatrix} -3 & 2 \\ -4 & 3 \\ -5 & 4 \end{bmatrix}. \quad (7)$$

Problem: Let $L : \mathbb{R}^2 \rightarrow \mathbb{R}^1$ be a linear transformation defined by

$$L\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = [x_1 + 3x_2] \quad (8)$$

Find the standard matrix representing L .

Solution: We want to find $L(\mathbf{e}_1)$ and $L(\mathbf{e}_2)$ by plugging in $x_1 = 1, x_2 = 0$ and $x_1 = 0, x_2 = 1$. This is equivalent to

$$L(\mathbf{e}_1) = L\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = [1] \quad (9)$$

$$L(\mathbf{e}_2) = L\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = [3] \quad (10)$$

Hence, we have $\begin{bmatrix} 1 & 3 \end{bmatrix}$.

- Know how to find the kernel and/or range/image of a linear transformation (as well as a basis for the kernel/image)

Definition: Given a linear transformation $T : V \rightarrow W$,

1. The kernel of L , denoted as $\ker T$, is the subset of V consisting of all elements $\mathbf{v} \in V$ such that $T(\mathbf{v}) = \mathbf{0}_W$.

2. The image (also called range) of T , denoted as $\text{im } T$, is the set of all vectors in W that are images under T of vectors in V , i.e. it is the set of $\mathbf{w} \in W$ such that there exists a vector $\mathbf{v} \in V$ such that $T(\mathbf{v}) = \mathbf{w}$.

Theorem: Given a linear transformation $T : V \rightarrow W$,

1. $\ker T$ is a subspace of V .
2. $\text{im } T$ is a subspace of W .

Theorem: Given a linear transformation $L : V \rightarrow W$,

$$\dim \ker L + \dim \text{im } L = \dim V. \quad (11)$$

Tip: For a linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$, finding the basis for $\ker T$ and a basis for $\text{im } T$ is equivalent to finding the basis for the nullspace of A and a basis for $\text{im } A$, respectively, where A is the standard matrix representing T .

One can take the following strategy to identify $\ker T$ for $T : V \rightarrow W$ as follows.

1. Set the image of an arbitrary vector equal to the zero vector $\mathbf{0}$, i.e. set $T(\mathbf{v}) = \mathbf{0}_W$.
2. Solve the resulting linear system.
3. Parametrize the solution using free variables.
4. Rewrite the parametrized solution back in the original form.

Here is a strategy to identify $\text{im } T$ as follows.

1. Use the equation $\dim \ker T + \dim \text{im } T = \dim V$ to figure out what the dimension of the image is.
2. Find a spanning set from the definition of the linear transformation T .
3. Find a basis for $\text{im } T$ by reducing the spanning set down to a linearly independent set (use the $\dim \text{im } T$).

Warning: Your final answers for basis of Kernel / Image / need to be in the correct vector space.

Problem: Consider the linear transformation $L : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ whose standard matrix representing L is given by

$$\begin{bmatrix} -3 & 2 \\ -4 & 3 \\ -5 & 4 \end{bmatrix}. \quad (12)$$

Show that L is one-to-one, and find the basis of the image of L .

Solution: We note that the RREF of the standard matrix representing L is given by

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}. \quad (13)$$

Hence, because there are no columns without pivots, it follows that L is one-to-one.

Recall that

$$0 + \dim \text{im } L = \dim \ker L + \dim \text{im } L = \dim V = 2. \quad (14)$$

Hence, $\dim L = 2$. Note that the set $S = \{L(\mathbf{e}_1), L(\mathbf{e}_2)\}$ is given by

$$S = \left\{ \begin{bmatrix} -3 \\ -4 \\ -5 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \right\}. \quad (15)$$

Because the number of elements of S is identical to the dimension of $\dim L$, we obtain that S is the basis of the image of L .

Problem: Consider the linear transformation $L : P_1 \rightarrow M_{2 \times 2}$ defined as

$$L(ax + b) = \begin{bmatrix} a - 2b & 0 \\ 4b - 2a & 0 \end{bmatrix} \quad (16)$$

Identify the basis of $\ker L$ and $\dim L$.

Solution: We notice that the kernel of L can be obtained from finding $ax + b$ such that

$$L(ax + b) = \begin{bmatrix} a - 2b & 0 \\ 4b - 2a & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad (17)$$

Therefore, we have

$$a - 2b = 0 \quad (18)$$

$$4b - 2a = 0 \quad (19)$$

Note that both equations are scalar multiples of one another, and one obtains the relation that $a = 2b$. Regarding b as a free variable, any polynomial in Kernel of L can be written as

$$ax + b = 2bx + b = b(2x + 1) \quad (20)$$

Therefore, the basis for the kernel of L is the set $\{2x + 1\}$.

Using the equation

$$1 + \dim \dim L = \dim \ker L + \dim \dim L = \dim P_1 = 2. \quad (21)$$

we obtain that $\dim \dim L = 1$. Hence, it suffices to find a set with a single matrix which spans the image of L . Observe that

$$\begin{bmatrix} a - 2b & 0 \\ 4b - 2a & 0 \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ -2 & 0 \end{bmatrix} + b \begin{bmatrix} -2 & 0 \\ 4 & 0 \end{bmatrix} \quad (22)$$

Therefore, the set $\left\{ \begin{bmatrix} 1 & 0 \\ -2 & 0 \end{bmatrix}, \begin{bmatrix} -2 & 0 \\ 4 & 0 \end{bmatrix} \right\}$ spans the image of L . But notice that $\begin{bmatrix} -2 & 0 \\ 4 & 0 \end{bmatrix}$ is a scalar multiple of $\begin{bmatrix} 1 & 0 \\ -2 & 0 \end{bmatrix}$ by 2. Therefore, the set $\left\{ \begin{bmatrix} 1 & 0 \\ -2 & 0 \end{bmatrix} \right\}$ is a basis of the image of L .

Problem: Let $L : P_3 \rightarrow P_3$ be the linear transformation given by

$$L(ax^3 + bx^2 + cx + d) = (c - d)x^3 + (a - b)x. \quad (23)$$

Find a basis for $\ker L$ and $\dim L$.

Solution:

In order for the element $ax^3 + bx^2 + cx + d$ to lie in the kernel of L , we require that

$$\begin{cases} c - d = 0 \\ a - b = 0 \end{cases} \quad (24)$$

These imply that $a = b$ and $c = d$, with taking b and d as free variables. This allows one to express the element as

$$ax^3 + bx^2 + cx + d = b(x^3 + x^2) + d(x + 1) \quad (25)$$

Hence, the set $\{x^3 + x^2, x + 1\}$ spans the kernel of L . Because the set is clearly linearly independent (as neither vector is a scalar multiple of the other), the set is a basis of the kernel of L .

Because we have

$$\dim \ker L + \dim \operatorname{im} L = \dim P_3 = 4, \quad (26)$$

it follows that $\dim \operatorname{im} L = 2$. Hence, a basis for $\operatorname{im} L$ is a set of two vectors. By observing how L is defined, we notice that the basis for image of L is $\{x^3, x\}$, because the image of L is forced to have the coefficients of x^2 and the constant term to be equal to 0.

- Know how to determine if a linear transformation is one-to-one and/or onto

Definition/Theorem: Let $T : V \rightarrow W$ be a linear transformation.

1. T is one-to-one (injective) if $T(\mathbf{v}) = \mathbf{0}_W$, then $\mathbf{v} = \mathbf{0}_V$, i.e. $\ker(T) = \{\mathbf{0}_V\}$.
 - Technically, the definition that T is one-to-one (injective) is: $T(\mathbf{v}_1) = T(\mathbf{v}_2)$ implies that $\mathbf{v}_1 = \mathbf{v}_2$. However, we showed in class that the above definition is equivalent (and is a way easier test for one-to-one!).
2. T is onto (surjective) if for every vector $\mathbf{w} \in W$, there exists a vector $\mathbf{v} \in V$ such that $f(\mathbf{v}) = \mathbf{w}$, i.e. $\operatorname{im} T = W$. Note: this also equivalent to showing that $\dim \operatorname{im} T = \dim W$

Definition: Given a linear transformation $T : V \rightarrow W$:

$\dim(\ker T)$ is called the nullity of T and is denoted $\text{nullity}(T)$.

$\dim(\operatorname{im} T)$ is called the rank of T and is denoted $\text{rank}(T)$.

Tip: A linear transformation $T : V \rightarrow W$ is one-to-one if and only if $\text{nullity}(T) = 0$.

Problem: Suppose $L : \mathbb{R}^3 \rightarrow \mathbb{R}^1$ is the linear transformation given as

$$L \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = [x + y + z]. \quad (27)$$

Verify that L is onto/surjective but not one-to-one, and compute the rank and nullity of L .

Solution:

Note that L is onto because for any number $a \in \mathbb{R}^1$, we have

$$L \left(\begin{bmatrix} a \\ 0 \\ 0 \end{bmatrix} \right) = [a] \quad (28)$$

On the other hand, L is not one-to-one. This is because we have

$$L \left(\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right) = [0] \quad (29)$$

and $\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ is not the zero vector.

The standard matrix representing L can be obtained from obtaining $L(\mathbf{e}_1)$, $L(\mathbf{e}_2)$, and $L(\mathbf{e}_3)$. In particular, we have

$$L(\mathbf{e}_1) = L \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right) = [1], \quad L(\mathbf{e}_2) = L \left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right) = [1], \quad L(\mathbf{e}_3) = L \left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right) = [1]. \quad (30)$$

Hence, the standard matrix of L is $\begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$.

Notice that L is already in RREF. We can regard y and z as two free variables, and $x = -y - z$. Hence, the nullity of L is equal to 2. The rank of L is equal to 1 because there is only one pivot in RREF of L . (And indeed, the Rank-Nullity theorem certifies that our answer is correct: $1 + 2 = 3$, the number of columns of L).

- Know the dimension theorem and how to use it to calculate possible dimensions of kernel and image.

Recall: (Dimension Theorem) Given a linear transformation $T : V \rightarrow W$,

$$\dim \ker T + \dim \operatorname{im} T = \dim V. \quad (31)$$

Problem: If $T : P_5 \rightarrow M_{2 \times 2}$ is a linear transformation, What's the smallest/largest dimension of $\ker T$ possible? What's the smallest/largest dimension of $\operatorname{im} T$ possible?

Solution: Since $V = P_5$ in this case, and $\dim P_5 = 6$, this means that $\dim \ker T + \dim \operatorname{im} T = 6$. Also, since $\operatorname{im} T$ is a subspace of $M_{2 \times 2}$, the largest the dimension $\operatorname{im} T$ can be is $\dim M_{2 \times 2} = 4$.

Is it possible to build a linear transformation $T : P_5 \rightarrow M_{2 \times 2}$ with $\dim \operatorname{im} T = 4$ (try it!), so that means that smallest the $\dim \ker T$ can be is 2, since $6 - 4 = 2$.

The largest $\dim \ker T$ possible is 6 (by taking $T : P_5 \rightarrow M_{2 \times 2}$ to be the zero transformation), and that would result in $\dim \operatorname{im} T = 0$.

- Know how the dimension of vector spaces for a linear transformation interacts with the problem of being one-to-one/onto

Recall: (Dimension Theorem) Given a linear transformation $T : V \rightarrow W$,

$$\dim \ker T + \dim \operatorname{im} T = \dim V. \quad (32)$$

Notice that this tells us several things about what's "possible" for a linear transformation:

- $\dim V > \dim W$, it's impossible for $T : V \rightarrow W$ to be one-to-one (Why?)
- If $\dim V < \dim W$ it's impossible for $T : V \rightarrow W$ to be onto (Why?)

Problem: If $T : P_5 \rightarrow M_{22}$ is a linear transformation...

- Is it possible for T to be one-to-one?
- Is it possible for T to be onto?

Solution:

(a) No, it's not possible to be one-to-one. If T is one-to-one, then $\dim \ker(T) = 0$, so using the Dimension Theorem, $0 + \dim \operatorname{im} T = \dim(P_5) = 6$. This would force $\dim \operatorname{im} T = 6$, but since $\operatorname{im} T$ is a subspace of a 4-dimensional vector space, this is impossible.

(b) It is possible for T to be onto. Notice it would require $\dim \ker(T) = 2$. An example of this is:

$$T(ax^5 + bx^4 + cx^3 + dx^2 + ex + f) = \begin{bmatrix} c & d \\ e & f \end{bmatrix}.$$

Notice that a linear transformation from a larger dimensional vector space to a smaller dimensional vector space is not required to be onto. For example, $T(ax^5 + bx^4 + cx^3 + dx^2 + ex + f) = \begin{bmatrix} c & d \\ 0 & 0 \end{bmatrix}$ is neither one-to-one nor onto.

Similarly, a linear transformation from a smaller dimensional vector space to a larger dimensional vector space is not required to be one-to-one.

- Be able to determine if a linear transformation is an isomorphism (both using the definition and using a matrix representation)

Definition: A linear transformation $T : V \rightarrow W$ is an isomorphism if it is both one-to-one and onto.

Theorem: If V and W have the same dimension n , a linear transformation $T : V \rightarrow W$ is an isomorphism if it is either one-to-one or onto.

Example: Is $T : P_3 \rightarrow M_{22}$ by $T(p(x)) = \begin{bmatrix} p(0) & p(1) \\ p(0) + p(1) & p(0) - p(1) \end{bmatrix}$ an isomorphism?

T linear: Let $p(x), q(x)$ be in P_3 , and a in $\mathbb{R}$. Then

$$\begin{aligned} T(p(x) + q(x)) &= \begin{bmatrix} p(0) + q(0) & p(1) + q(1) \\ p(0) + q(0) + p(1) + q(1) & p(0) + q(0) - (p(1) + q(1)) \end{bmatrix} \\ &= \begin{bmatrix} p(0) & p(1) \\ p(0) + p(1) & p(0) - p(1) \end{bmatrix} + \begin{bmatrix} q(0) & q(1) \\ q(0) + q(1) & q(0) - q(1) \end{bmatrix} \\ &= T(p(x)) + T(q(x)) \end{aligned}$$

$$\begin{aligned} T(ap(x)) &= \begin{bmatrix} ap(0) & ap(1) \\ ap(0) + ap(1) & ap(0) - ap(1) \end{bmatrix} \\ &= a \begin{bmatrix} p(0) & p(1) \\ p(0) + p(1) & p(0) - p(1) \end{bmatrix} \\ &= aT(p(x)) \end{aligned}$$

Is T one-to-one? If $T(p(x)) = \begin{bmatrix} p(0) & p(1) \\ p(0) + p(1) & p(0) - p(1) \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, does this imply $p(x) = 0$?

Nope, consider: $p(x) = x(x-1)$. $p(0) = 0 = p(1)$, so $T(p(x)) = \begin{bmatrix} p(0) & p(1) \\ p(0) + p(1) & p(0) - p(1) \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, but $p(x)$ isn't the zero polynomial.

So T has a non-trivial kernel, so T is not one-to-one. Since P_3 and M_{22} both have dimension 4, this means T is also automatically not onto.

Note that P_3 and M_{22} are isomorphic vector spaces, but this particular T is not an isomorphism.

Theorem Let V and W be finite dimensional vector spaces. The following conditions are equivalent for a linear transformation $T : V \rightarrow W$.

1. T is an isomorphism.
2. There exists a linear transformation $S : W \rightarrow V$ such that $ST = 1_V$ and $TS = 1_W$

In this case, S is called the inverse of T and is denoted by T^{-1} .

We can also use a matrix representation of a linear transformation to determine if the linear transformation is an isomorphism:

Theorem: Let $T : V \rightarrow W$ be a linear transformation, where $\dim V = \dim W = n$. The following are equivalent.

1. T is an isomorphism.
2. $M_{DB}(T)$ is invertible for all ordered bases B and D of V and W .

When this is the case, $[M_{DB}(T)]^{-1} = M_{BD}(T^{-1})$.

Example: If $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) = \begin{bmatrix} -2x_1 + 3x_2 \\ 4x_1 - 6x_2 \end{bmatrix}$, we can use the standard basis $\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$ for both B and D to build the standard matrix representation of T : $A = M_{DB}(T) = \begin{bmatrix} -2 & 3 \\ 4 & -6 \end{bmatrix}$. Since $\det(A) = (-2)(-6) - (3)(4) = 0$, A is not invertible, so T_A is not invertible and is not an isomorphism.

As another example, $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) = \begin{bmatrix} 2x_1 + 3x_2 \\ x_1 + 3x_2 \end{bmatrix}$, we can use the standard basis $\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$ for both B and D to build the standard matrix representation of T : $B = M_{DB}(T) = \begin{bmatrix} 2 & 3 \\ 1 & 3 \end{bmatrix}$, then since $\det(B) = (2)(3) - (1)(3) = 3 \neq 0$, B is invertible, so T is invertible. We can compute B^{-1} (either by using the 2×2 formula or by reducing $[B : I_2]$) to find $B^{-1} \begin{bmatrix} 1 & -1 \\ -1/3 & 2/3 \end{bmatrix}$. So $T^{-1} \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) = B^{-1} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 - x_2 \\ -\frac{1}{3}x_1 + \frac{2}{3}x_2 \end{bmatrix}$.

- Know how to determine if two finite-dimensional vector spaces are isomorphic

Definition: V and W are isomorphic if there exists an isomorphism $T : V \rightarrow W$.

Tip: Two vector spaces V and W are isomorphic if and only if $\dim V = \dim W$.

Example: $\mathbb{R}^4$ and M_{22} and P_3 are isomorphic, because all these have dimensions equal to 4.

Problem: For which d is P_d isomorphic to $W =$ the set of 2×2 symmetric matrices?

Solution: The dimension of W is 3 (you can show that a basis is $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$), so since $\dim P_d = d + 1$, we need $d = 2$ for the dimensions to match.

- Be able to give the coordinate vector with respect to an ordered basis

Definition: Let $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ be a basis in V . We call $B = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ an ordered basis of V if the order of the vectors is fixed. (i.e. $B_1 = (\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n)$ and $S_2 = (\mathbf{b}_2, \mathbf{b}_1, \dots, \mathbf{b}_n)$ are different ordered basis).

Tip: given an ordered basis $B = (\mathbf{b}_1, \dots, \mathbf{b}_n)$, we can represent any vector $\mathbf{v}$ in $\text{Span} B$ as

$$\mathbf{v} = a_1 \mathbf{b}_1 + \dots + a_n \mathbf{b}_n \quad (33)$$

and represent in column vector as

$$C_B(\mathbf{v}) := \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}. \quad (34)$$

Example: Given an ordered basis $B = (x^2 + 2, x + 1, 3)$ of P_2 , the polynomial $p = x^2 + x + 2$ can be written as

$$x^2 + x + 2 = 1 \cdot (x^2 + 2) + 1 \cdot (x + 1) + \left(-\frac{1}{3}\right) \cdot 3. \quad (35)$$

Hence, as a column vector, we can write p as

$$C_B(p) = \begin{bmatrix} 1 \\ 1 \\ -\frac{1}{3} \end{bmatrix}. \quad (36)$$

- Know how to find the matrix representing a linear transformation $T : V \rightarrow W$ using ordered bases for vector spaces V and W

Strategy: Suppose $T : V \rightarrow W$ is a linear transformation, with $B = (v_1, v_2, \dots, v_n)$ an ordered basis of V and $D = (w_1, \dots, w_m)$ an ordered basis of W . We can find the matrix representing T as follows.

1. Find the outputs of vectors in B , i.e. compute $T(v_1), T(v_2), \dots, T(v_n)$.
2. Express the outputs from Step (1) as a linear combination using the ordered basis D . For example, suppose we have the following n equations: (we indicate the different linear combinations in different colors for the ease of exposition in Step (3)).

$$\begin{aligned} T(v_1) &= \mathbf{a_{11}}w_1 + \mathbf{a_{21}}w_2 + \dots + \mathbf{a_{m1}}w_m \\ T(v_2) &= \mathbf{a_{12}}w_1 + \mathbf{a_{22}}w_2 + \dots + \mathbf{a_{m2}}w_m \\ &\vdots \\ T(v_n) &= \mathbf{a_{1n}}w_1 + \mathbf{a_{2n}}w_2 + \dots + \mathbf{a_{mn}}w_m \end{aligned} \quad (37)$$

3. Write the coefficients obtained from Step (2) into a matrix. In particular, the $\mathbf{i}$ -th output $\mathbf{T(v_i)}$ gives you the $\mathbf{i}$ -th **column** of the matrix. Using the colored entries from Step (2), we can write the matrix as

$$\begin{bmatrix} \mathbf{a_{11}} & \mathbf{a_{12}} & \dots & \mathbf{a_{1n}} \\ \mathbf{a_{21}} & \mathbf{a_{22}} & \dots & \mathbf{a_{2n}} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{a_{m1}} & \mathbf{a_{m2}} & \dots & \mathbf{a_{mn}} \end{bmatrix} \quad (38)$$

Note: If $T : V \rightarrow V$, we can represent T as a matrix using the same basis B for the “beginning” and “end”: $M_B(T) = M_{BB}(T)$ is called the B -matrix of T .

Example: Suppose $T : P_1 \rightarrow P_1$ with $T(ax + b) = (a + b)x + a + 2b$. Let’s suppose B and D are the following ordered bases

$$\begin{aligned} B &= (x, 1) \\ D &= (x - 2, 1) \end{aligned} \quad (39)$$

Let’s compute $M_{DB}(T)$.

$$\begin{aligned} T(x) &= x + 1 = \mathbf{1} \cdot (x - 2) + \mathbf{3} \cdot 1 \\ T(1) &= x + 2 = \mathbf{1} \cdot (x - 2) + \mathbf{4} \cdot 1 \end{aligned} \quad (40)$$

Hence, the matrix representing T with respect to ordered bases B and D is:

$$M_{DB}(T) = \begin{bmatrix} \mathbf{1} & \mathbf{1} \\ \mathbf{3} & \mathbf{4} \end{bmatrix} \quad (41)$$

Let's compute $M_{BD}(T)$.

$$\begin{aligned} T(x-2) &= -x-3 = -\mathbf{1} \cdot (x) + -\mathbf{3} \cdot 1 \\ T(1) &= x+2 = \mathbf{1} \cdot (x) + \mathbf{2} \cdot 1 \end{aligned} \quad (42)$$

Hence, the matrix representing T with respect to ordered bases D and B is:

$$M_{DB}(T) = \begin{bmatrix} -\mathbf{1} & \mathbf{1} \\ -\mathbf{3} & \mathbf{2} \end{bmatrix} \quad (43)$$

Let's compute $M_B(T) = M_{BB}(T)$.

$$\begin{aligned} T(x) &= x+1 = \mathbf{1} \cdot (x) + \mathbf{1} \cdot 1 \\ T(1) &= x+2 = \mathbf{1} \cdot (x) + \mathbf{2} \cdot 1 \end{aligned} \quad (44)$$

Hence, the B -matrix of T is:

$$M_B(T) = \begin{bmatrix} \mathbf{1} & \mathbf{1} \\ \mathbf{1} & \mathbf{2} \end{bmatrix} \quad (45)$$

Let's compute $M_D(T) = M_{DD}(T)$.

$$\begin{aligned} T(x-2) &= -x-3 = -\mathbf{1} \cdot (x-2) + -\mathbf{5} \cdot 1 \\ T(1) &= x+2 = \mathbf{1} \cdot (x-2) + \mathbf{4} \cdot 1 \end{aligned} \quad (46)$$

Hence, the D -matrix of T is:

$$M_D(T) = \begin{bmatrix} -\mathbf{1} & \mathbf{1} \\ -\mathbf{5} & \mathbf{4} \end{bmatrix} \quad (47)$$

Warning: When computing the image of a linear transformation $T(\mathbf{v})$ using a matrix representation $M_{DB}(T)$, it is crucial to follow the following steps:

1. Write the vector $\mathbf{v}$ as a coordinate vector $C_B(\mathbf{v})$ **with respect to an ordered basis B given in the problem.**
2. Multiply the matrix $M_{DB}(T)$ with $C_B(\mathbf{v})$: $M_{DB}(T)C_B(\mathbf{v}) = C_D(T(\mathbf{v}))$
3. Re-express the column vector $C_D(T(\mathbf{v}))$ as an element in the vector space W **using the ordered basis D given in the problem.**

Let's return back to the previous example for $T : P_1 \rightarrow P_1$ with $T(ax+b) = (a+b)x + a + 2b$. Suppose we want to compute $L(4x+2)$. If we use $M_{DB}(T)$, the procedure for obtaining $L(4x+2)$ is as follows:

Step 1: Express $4x+2$ as a column vector with respect to $B = (x, 1)$.

Note that we have $4x+2 = 4 \cdot x + 2 \cdot 1$.

$$C_B(4x+2) = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

Step 2: Matrix multiplication using $M_{DB}(T)$.

$$\begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 20 \end{bmatrix}$$

Step 3: Re-express the column vector as an element of P_1 using $D = (x-2, 1)$.

$$C_D^{-1} \left(\begin{bmatrix} 6 \\ 20 \end{bmatrix} \right) = 6 \cdot (x-2) + 20 \cdot (1) = 6x+8$$

- Be able to find kernel/image of a linear transformation using a matrix representation

theorem: Let $T : V \rightarrow W$ be a linear transformation, with ordered bases B and D of V and W , respectively. Then:

1. A vector $\mathbf{v}$ in V is in $\ker T$ if and only if $C_B(\mathbf{v})$ is in the null space of $M_{DB}(T)$.
2. The dimension of the kernel of T is equal to dimension of the null space of $M_{DB}(T)$.
3. A vector $\mathbf{w}$ in W is in $\text{im } T$ if and only if $C_D(\mathbf{w})$ is in the column space of $M_{DB}(T)$.
4. The dimension of the image of T is equal to the dimension of the column space of $M_{DB}(T)$.

To find the kernel/image of a linear transformation T using a matrix representation $M_{DB}(T)$, we find a basis for the nullspace/image space of the $M_{DB}(T)$, then “translate” it back to the correct vector space using the coordinate isomorphism.

Problem: Consider $T : M_{2 \times 2} \rightarrow P_2$ by $T \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) = (a + 2b)x^2 + (c - d)x - (a + 2b)$. Verify on your own that this is a linear transformation. Use a matrix representation to find a basis for the kernel and image of T .

Solution: To find the kernel/image, we build a matrix representation of T . I choose $B = \left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right)$ and $D = (x^2, x, 1)$ for my ordered bases. Now I compute the image of each vector in B :

$$T \left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \right) = x^2 - 1$$

$$T \left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \right) = 2x^2 - 2$$

$$T \left(\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \right) = x$$

$$T \left(\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right) = -x$$

I convert each image into its coordinate representation using D , then build $M_{DB}(T) = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ -1 & -2 & 0 & 0 \end{bmatrix}$.

$\begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ -1 & -2 & 0 & 0 \end{bmatrix}$ reduces to $\begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. To compute the nullspace, we note that $x_2 = s$ and $x_4 = t$ are free variables with $x_1 + 2x_2 = 0$ and $x_3 - x_4 = 0$, so our solutions look like

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -2s \\ s \\ t \\ t \end{bmatrix} = s \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}.$$

$\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right\}$ is a basis for the nullspace of $M_{DB}(T)$, so we translate back to $M_{2 \times 2}$ using C_B^{-1} :

$$C_B^{-1} \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 0 & 0 \end{bmatrix}$$

$$C_B^{-1} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}, \text{ so } \left\{ \begin{bmatrix} -2 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \right\} \text{ is a basis for the kernel of } T.$$

The pivots of RREF of $M_{DB}(T)$ are in columns 1 and 3, so columns 1 and 3 of $M_{DB}(T)$ are a basis for the image space: $\left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$. Now translate back to P_2 using C_D^{-1} :

$$C_D^{-1} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = x^2 - 1$$

$$C_D^{-1} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = x$$

This means that $\{x^2 - 1, x\}$ is a basis for the image of T .

- Know how to change the representation of a linear operator $T : V \rightarrow V$ from one basis to another, both directly and using the Similarity Theorem

Definition: Let $B = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ and D be two ordered basis of an n -dimensional real vector space V .

Then the transition matrix (also called the change matrix) $P_{D \leftarrow B}$ from the B -basis to the D -basis is given by

$$P_{D \leftarrow B} = [C_D(\mathbf{b}_1) \ C_D(\mathbf{b}_2) \ \cdots \ C_D(\mathbf{b}_n)] \quad (48)$$

Note: The transition matrix is a special case of a matrix representation: $P_{D \leftarrow B} = M_{DB}(1_V)$.

Tip: We in fact have the relation

$$C_D(\mathbf{v}) = P_{D \leftarrow B} C_B(\mathbf{v}) \quad (49)$$

Tip: By construction, we have

$$P_{D \leftarrow B}^{-1} = P_{B \leftarrow D} \quad (50)$$

Note: The Similarity Theorem is only for Matrix Representations of linear operators $T : V \rightarrow V$ where the two ordered bases are the same: $M_B(T) = M_{BB}(T)$.

(Similarity Theorem) Let B and D be two ordered bases of a finite dimensional vector space V . If $T : V \rightarrow V$ is any linear operator, the matrices $M_B(T)$ and $M_D(T)$ of T with respect to these bases are similar. More precisely,

$$M_B(T) = P^{-1} M_D(T) P$$

where $P = P_{D \leftarrow B}$ is the transition matrix from B to D .

In other words,

$$M_B(T) = P_{B \leftarrow D} M_D(T) P_{D \leftarrow B}.$$

(Note how the arrows line up in the formula).

Example Let's verify the Similarity Theorem using $M_B(T)$ and $M_D(T)$ from our previous example: $T : P_1 \rightarrow P_1$ with $T(ax + b) = (a + b)x + a + 2b$ and $B = (x, 1), D = (x - 2, 1)$.

We need to compute $P = P_{D \leftarrow B}$ and P^{-1} .

$$P_{D \leftarrow B} = [C_D(x) \ C_D(1)]$$

$$x = 1(x - 2) + 2(1), \text{ so } C_D(x) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$1 = 0(x - 2) + 1(1), \text{ so } C_D(1) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\text{So } P_{D \leftarrow B} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}.$$

$$P^{-1} = \frac{1}{\det(P)} \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$$

$$\text{Now we compute } P^{-1}M_D(T)P = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ -5 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} = M_B(T)$$

- Be able to find the eigenvalues/eigenvectors of a linear operator

Definition: A real number λ is called an eigenvalue of an operator $T : V \rightarrow V$ if $T(\mathbf{v}) = \lambda\mathbf{v}$ for some nonzero vector $\mathbf{v}$ in V . In this case, $\mathbf{v}$ is called an eigenvector of T corresponding to λ .

Tip: To find the eigenvalues/eigenvectors for a general linear operator $T : V \rightarrow V$:

1. Pick any ordered basis B for V .
2. Build the matrix $M_B(T)$.
3. Compute the eigenvalues and eigenvectors for $M_B(T)$
4. T has the same eigenvalues. To get the eigenvectors, “translate” back to V from $\mathbb{R}^n$ using C_B^{-1} .

Example: Consider $L : P_2 \rightarrow P_2$ by $L(p(x)) = p''(x) + 2p'(x) + p(1)x$.

I choose $B = (1, x, x^2)$ to build $M_B(T)$.

$$L(1) = x \longrightarrow C_B(1) = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$L(x) = 2 + x \longrightarrow C_B(1) = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

$$L(x^2) = 2 + 5x \longrightarrow C_B(1) = \begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix}$$

$$M = M_B(L) = \begin{bmatrix} 0 & 2 & 2 \\ 1 & 1 & 5 \\ 0 & 0 & 0 \end{bmatrix}$$

Now we find the eigenvalues and eigenvectors for M :

$$\det(\lambda I - M) = \det \begin{bmatrix} \lambda & -2 & -2 \\ -1 & \lambda - 1 & -5 \\ 0 & 0 & \lambda \end{bmatrix} = \lambda(\lambda(\lambda - 1) - 2) = \lambda(\lambda^2 - \lambda - 2) = \lambda(\lambda - 2)(\lambda + 1) = 0$$

So our eigenvalues are $\lambda = 0, 2, -1$.

You can compute a basis for each eigenspace (this left as an exercise):

$$\lambda = 0: \begin{bmatrix} -4 \\ -1 \\ 1 \end{bmatrix}$$

$$\lambda = -1: \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

$$\lambda = 2: \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

Now to get our eigenvectors for L , we “translate” back to P_2 using our basis B :

$$\lambda = 0: C_B(\mathbf{v}) = \begin{bmatrix} -4 \\ -1 \\ 1 \end{bmatrix} \longrightarrow \mathbf{v} = -4 - x + x^2$$

$$\lambda = -1: C_B(\mathbf{w}) = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \longrightarrow \mathbf{w} = -2 + x$$

$$\lambda = 2: C_B(\mathbf{u}) = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \longrightarrow \mathbf{u} = 1 + x$$

Note: we can check our work! Let’s compute the image of each eigenvector using L :

$$\lambda = 0: L(\mathbf{v}) = L(x^2 - x - 4) = 2 + 2(2x - 1) - (1 - 1 - 4)x = 2 + 4x - 2 - 4x = 0 = 0(x^2 - x - 4)$$

$$\lambda = -1: L(\mathbf{w}) = L(x - 2) = 0 + 2(1) + (-1)x = -x + 2 = -1(x - 2)$$

$$\lambda = 2: L(\mathbf{u}) = L(x + 1) = 0 + 2(1) + 2x = 2(x + 1)$$

- Be able to find a diagonal matrix representation of a linear operator (if possible)

Theorem: Let $T : V \rightarrow V$ be a linear operator on a finite dimensional space V . Then the following conditions are equivalent.

1. V has a basis consisting of eigenvectors of T .
2. There exists a basis B of V such that $M_B(T)$ is diagonal.

If either case holds, T is called diagonalizable.

Tip: T is diagonalizable exactly when a matrix $M_B(T)$ representing T is diagonalizable.

Example: $L : P_2 \rightarrow P_2$ by $L(p(x)) = p''(x) + 2p'(x) + p(1)x$ is diagonalizable because the matrix $M_B(T)$ for $B = (1, x, x^2)$ is diagonalizable (you can see this because it has three distinct eigenvalues).

We found corresponding eigenvectors for each eigenvalue in the previous example, so we can use these to build a basis for P_2 : $U = (x^2 - x - 4, x - 2, x + 1)$ (why are they a basis)?

$M_U(L)$ will be a diagonal matrix (note: This is because I am using the eigenvectors of L for my basis U):

$$L(x^2 - x - 4) = 2 + 2(2x - 1) - (1 - 1 - 4)x = 2 + 4x - 2 - 4x = 0 = 0(x^2 - x - 4) \rightarrow C_U(0) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$L(x - 2) = 0 + 2(1) + (-1)x = -x + 2 = -1(x - 2) \rightarrow C_U(-x + 2) = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$$

$$L(x+1) = 0 + 2(1) + 2x = 2(x+1) \rightarrow C_U(2x+2) = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

$$\text{So } M_U(L) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

- Know how to verify if a function is an inner product

Definition: Given a vector space V , an inner product on V is a function $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ which satisfies the following properties for any vectors $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$:

1. $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$.
2. $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$
3. $\langle c\mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, c\mathbf{v} \rangle = c\langle \mathbf{u}, \mathbf{v} \rangle$ for any real scalar c .
4. $\langle \mathbf{u}, \mathbf{u} \rangle > 0$ for all $\mathbf{u} \neq \mathbf{0}$.

(Note: $V \times V \rightarrow \mathbb{R}$ means that you take a pair of vectors in V and assign to them a real number)

Non-Example: The function $F : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ defined as

$$F\left(\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}, \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}\right) = v_1 w_1 - v_2 w_2 \quad (51)$$

is NOT an inner product of $\mathbb{R}^2$, because $F\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = 0$ whereas $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is not the zero vector of $\mathbb{R}^2$.

Example: Recall that P_d is the set of polynomials in x of degree less than or equal to d . The function $\langle \cdot, \cdot \rangle : P_d \times P_d \rightarrow \mathbb{R}$ defined as

$$\langle f(x), g(x) \rangle := \int_0^1 f(x)g(x)dx \quad (52)$$

is an inner product of P_d .

- Be able to compute the length/norm of a vector and find a unit vector given an inner product

Tip: Similar to standard inner product, the length/norm of a vector $\mathbf{v}$ in a vector space V with an inner product $\langle \cdot, \cdot \rangle$ is

$$\|\mathbf{v}\| := \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle} \quad (53)$$

[[Unit Vector]] Given a vector $\mathbf{v}$, the **unit vector** $\mathbf{u}$ in the direction of $\mathbf{v}$ is

$$\mathbf{u} := \frac{1}{\|\mathbf{v}\|} \mathbf{v}. \quad (54)$$

Example: Recall the inner product defined over the set of polynomials P_d . Using P_1 , the length/norm of the polynomial $p(x) = x + 1$ is equal to

$$\|p(x)\| = \sqrt{\int_0^1 (x+1)^2 dt} = \sqrt{\int_0^1 x^2 + 2x + 1 dt} = \sqrt{\left[\frac{x^3}{3} + x^2 + x\right]_0^1} = \sqrt{\frac{7}{3}}. \quad (55)$$

The unit vector $u(x)$ in the direction of $p(x)$ is

$$u(x) = \frac{1}{\|p(x)\|} p(x) = \frac{1}{\sqrt{7/3}} (x+1) = \frac{\sqrt{3}}{\sqrt{7}} (x+1). \quad (56)$$

- Be able to find the distance between two vectors using an inner product

Definition: The distance between two vectors $\mathbf{u}$ and $\mathbf{v}$ in an inner product space is

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|.$$

Example: Using the inner product $\langle f(x), g(x) \rangle := \int_0^1 f(x)g(x)dx$ for P_1 , find the distance between $p(x) = 4x + 5$ and $q(x) = -2x + 3$

We compute $p(x) - q(x) = 6x - 2$. Then we find the norm of $6x - 2$:

$$\|6x - 2\| = \sqrt{\int_0^1 (6x - 2)^2 dx} = \sqrt{\int_0^1 36x^2 - 24x + 4 dx} = \sqrt{\left[\frac{36x^3}{3} - \frac{24x^2}{2} + 4x \right]_0^1} = 2$$

- Be familiar with the Cauchy-Bunyakovsky-Schwarz (CBS) inequality and its implications

Tip: The Cauchy-Bunyakovsky-Schwarz inequality still holds for any inner product of a vector space V :

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\| \quad (57)$$

As for the standard inner product spaces, we can define the angle θ between two vectors $\mathbf{u}, \mathbf{v} \in V$ as

$$\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|} \quad (58)$$

and the triangle inequality also holds.

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|. \quad (59)$$

Example: For the inner product defined over the set of polynomials P_d , the CBS inequality can be expressed as

$$\left| \int_0^1 f(x)g(x) dx \right| \leq \sqrt{\int_0^1 f(x)^2 dx} \sqrt{\int_0^1 g(x)^2 dx} \quad (60)$$

- Know how to check if two vectors are orthogonal given an inner product

[[Orthogonality]]: Two vectors $\mathbf{u}, \mathbf{v} \in V$ with an inner product $\langle \cdot, \cdot \rangle$ are orthogonal if and only if

$$\langle \mathbf{u}, \mathbf{v} \rangle = 0. \quad (61)$$

Example: Recall the inner product defined over the set of polynomials P_d . The two polynomials $p(x) = \sqrt{3}x + 1$ and $q(x) = \sqrt{3}x - 1$ are orthogonal with respect to the inner product because

$$(p(x), q(x)) := \int_0^1 (\sqrt{3}x + 1)(\sqrt{3}x - 1) dx = \int_0^1 3x^2 - 1 dx = [x^3 - x]_0^1 = 0. \quad (62)$$

- Be able to build an orthonormal basis given an orthogonal basis

Tip: If $\{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ is an orthogonal basis, then all we need to do to make this an orthonormal basis is divide each vector by its norm/length:

$\left\{ \frac{1}{\|\mathbf{b}_1\|} \mathbf{b}_1, \frac{1}{\|\mathbf{b}_2\|} \mathbf{b}_2, \dots, \frac{1}{\|\mathbf{b}_n\|} \mathbf{b}_n \right\}$ is an orthonormal basis.

Example: Verify that $S = \left\{ \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}$ is an orthogonal basis for $\mathbb{R}^2$ (using the standard inner product/dot product), and find an orthonormal basis using S .

We check that these vectors are orthogonal: $\begin{bmatrix} 1 \\ -2 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} = (1)(2) + (-2)(1) = 0$. This automatically makes them linearly independent, so S is an orthogonal basis for $\mathbb{R}^2$.

The length of both vectors is $\sqrt{1^2 + 2^2} = \sqrt{5}$, so

$$N = \left\{ \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\} = \left\{ \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{-2}{\sqrt{5}} \end{bmatrix}, \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix} \right\} \text{ is an orthonormal basis for } \mathbb{R}^2.$$

- Know how to find the coefficients of a linear combination using an orthogonal/orthonormal basis

Theorem: Let $S = \{\mathbf{f}_1, \dots, \mathbf{f}_n\}$ be an orthogonal basis for an inner product space V and let $\mathbf{v}$ be any vector in V . Then

$$\mathbf{v} = c_1 \mathbf{f}_1 + c_2 \mathbf{f}_2 + \dots + c_n \mathbf{f}_n$$

where

$$c_i = \frac{\langle \mathbf{v}, \mathbf{f}_i \rangle}{\|\mathbf{f}_i\|^2}$$

Example: Continuing our previous example from before, $\left\{ \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{-2}{\sqrt{5}} \end{bmatrix}, \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix} \right\}$ is an orthogonal basis for $\mathbb{R}^2$. Even better, it's an orthonormal basis, so when using this theorem, the denominator of each scalar is 1.

So if I want to find the linear combination of $\begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{-2}{\sqrt{5}} \end{bmatrix}$ and $\begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}$ to create $\mathbf{v} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$, I just take the dot product of $\mathbf{v}$ with each basis vector:

$$\begin{bmatrix} 3 \\ -2 \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{-2}{\sqrt{5}} \end{bmatrix} = \frac{3}{\sqrt{5}} + \frac{4}{\sqrt{5}} = \frac{7}{\sqrt{5}}$$

$$\begin{bmatrix} 3 \\ -2 \end{bmatrix} \cdot \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix} = \frac{6}{\sqrt{5}} - \frac{2}{\sqrt{5}} = \frac{4}{\sqrt{5}}$$

So this means

$$\begin{bmatrix} 3 \\ -2 \end{bmatrix} = \frac{7}{\sqrt{5}} \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{-2}{\sqrt{5}} \end{bmatrix} + \frac{4}{\sqrt{5}} \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}$$

- Be familiar with the list of invertibility criteria (Box of Facts)

The following statements are equivalent for an $n \times n$ matrix A .

1. A is nonsingular/invertible.
2. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
3. A is row equivalent to I_n .
4. The linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $n \times 1$ matrix $\mathbf{b}$.
5. $\text{rank}(A) = n$.
6. A can be expressed as a product of some number of elementary matrices.
7. $\det(A) \neq 0$.
8. The rows of A form a linearly independent set of n vectors in $\mathbb{R}^n$.
9. The columns of A form a linearly independent set of n vectors in $\mathbb{R}^n$.
10. The rows of A span $\mathbb{R}^n$.
11. The columns of A span $\mathbb{R}^n$.
12. The dimension of the nullspace of A is zero.
13. The dimension of the image space/column space of A is n .
14. The scalar $\lambda = 0$ is not an eigenvalue of A .
15. The linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ defined by $T(\mathbf{x}) = A\mathbf{x}$ for $\mathbf{x} \in \mathbb{R}^n$ is one-to-one.
16. The linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ defined by $T(\mathbf{x}) = A\mathbf{x}$ for $\mathbf{x} \in \mathbb{R}^n$ is onto.
17. The linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ defined by $T(\mathbf{x}) = A\mathbf{x}$ for $\mathbf{x} \in \mathbb{R}^n$ is an isomorphism.